

# Testing Weak Form Efficiency of Asian Markets: An Empirical Evaluation

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## Abstract

Why do we care whether stock market is efficient or not? Stock market acts as an intermediary to channelize funds from potential investors to firms who invest funds in projects to earn profit. Efficient market is a necessary prerequisite when it is desired that funds should be allocated to the highest-valued projects. Extant literature is dominated by the idea that Asian markets are efficient in weak form. The present paper attempts to examine the relevancy of efficient market hypothesis in five Asian markets for a period of fourteen years including the financial crisis. To test the weak form of market efficiency, the study analyses the pattern of short-term movements in the stock market returns and attempts to identify the underlying process generating those returns. Thus if any pattern is observed then the market is said to be inefficient and it can be inferred that the returns do not follow a random walk process. The study provides empirical evidence that Asian markets are not weak form efficient during the fourteen year period including financial crisis.

JEL Classification: G10, G14, G15

## Background

To the extent that capital markets are efficient, it makes it easier for the firm to raise capital as the market performs the price discovery process i.e. it determines the price at which market players are willing to buy claims on firm's future cash flows. This is possible only if stock prices are efficiently priced i.e. reflect the fundamental value of future discounted cash flows. When the general perception prevailing in the market is that prices accurately reflect information, participation cost will be low and stock market will successfully carry out its function of channelling resources to productive projects. Extant literature is dominated by the idea that Asian markets are efficient in weak form. This gives us the opportunity to explore this phenomenon with more stringent tests. Fama has talked about three types of efficiency weak, semi strong and strong. However, these efficiencies are on continuum, which means that initially weak form of efficiency is achieved thereafter semi strong and finally strong form

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is achieved. Therefore, firstly weak form efficiency needs to be achieved then subsequently we can talk about further efficiencies. Previous studies mostly stress that weak form of efficiency is achieved in Asian markets, but it have also been witnessed particularly during crisis period that markets get swept away.

### **Rationale of the Study**

If markets are swaying away particularly to one direction without any fundamental reason then efficiency hypothesis need to be examined. Therefore, in the present study we examine the relevancy of efficient market hypothesis in Asian markets. For this purpose we have selected five Asian countries. Asian region majorly consists of emerging economies, therefore we select four emerging economies and one developed economy. The five Asian countries were selected for analysis on the basis of GDP at purchasing power parity 2010. The selected countries are amongst top 12 in Asia in terms of GDP at purchasing power parity. If in our study we are able to conclude that markets are not efficient in weak form then it would indicate the fact that markets are not operating at its fundamental level. This implies that there exists opportunities for making profit but this opportunity can only be captured by big players and as a consequence markets are not good for small investors. In our study we shall examine that whether past prices are significant in explaining future prices through general linear regression model. Previous studies has been conducted by using special models like Runs test, Unit Root test, Variance Ratio test etc but these models have certain limitations and can be used in particular circumstances however we are conducting our analysis by using general regression model which is the most powerful general test.

### **Review of Literature**

The extent of literature is voluminous with respect to the studies concerning investigation of capital market efficiency in individual (both developed and developing) as well as in group of countries. Researchers have used different test procedures to examine the efficient market hypothesis and reported contradictory results. Some studies accepted the weak form efficiency while others refuted it. Thus, the controversy on the efficiency of emerging and developed capital markets continues over a period of time. Therefore, it is imperative to re-examine the issue in context of the financial crisis as well. Historically EMH is closely linked with random walk model which firstly was proposed by a French mathematician Louis Bachelier (1900) wherein he proposed that past, present and discounted future events are reflected in market price, but often show no apparent relation to price fluctuations. He also talked about informational efficiency and explained that if the market does not predict its fluctuations but is able to assess them as being more or less likely, and this likelihood can be evaluated mathematically. With a better understanding of price formation in competitive markets, the random walk model came to be seen as a set of observations that can be

consistent with the efficient markets hypothesis. Supporting evidence from other time series was provided by Working (1934) and from U.S. stock prices by Cowles and Jones (1937). These studies found that the serial correlation between successive price changes was essentially zero. Friedman (1953) explained that the EMH can be made only in the situations where the trading strategies of investors are correlated. Kendall (1953) analysed 22 price-series at weekly intervals and found to that they were essentially random. Also, he was the first to analyse time dependence of the empirical variance (non-stationarity).

In the early 1960s efficient market hypothesis was developed by Professor Eugene Fama, as an academic subject in his PhD thesis and published as characteristics of financial market. He suggested the first definition of the efficient market in 1969 and suggested that an efficient market is the market which adjusts rapidly to new information. This theory eventually became widely accepted theory of stock market movements. In developed markets, Fama (1965) empirically examined 30 stocks listed in the Dow Jones Industrial Average (DJIA) from 1956 to 1962. The findings suggested slight positive correlation which was statistically not different from zero and performed the run test which supported the findings, where in number of actual runs were less than expected. Fama concluded the efficiency of DJIA in the weak-form. Fama (1970) summarises the early random walk literature, his own contributions and other studies contained in the historical sequence of prices concludes that "the results are strongly in support" of the weak form of market efficiency. He concedes, however, that "much remains to be done", and indeed, Fama (1991) subsequently returned with a reinterpretation of the efficient markets hypothesis in the light of subsequent research. Jensen (1978) defines market efficiency as a market is said to be efficient with respect to information set  $\theta_t$ , if it is impossible to make economic profits by trading on the basis of information set  $\theta_t$ . According to him, the efficient market is where prices of securities reflect information up to the point in such a way that marginal benefits of acting on the information don't exceed the marginal cost of collecting it.

De Bondt and Thaler, 1985 discovered that stock prices overreact; evidencing substantial weak form market inefficiencies. Malkiel (1992) provided another definition of market efficiency. According to him a capital market is said to be efficient if it fully and correctly reflects all relevant information in determining security prices. According to him formally the market is said to be efficient with respect to some information set  $\Omega_1$ . Three points were emphasized in his definitions, first, the importance of the information set adopted in the test  $\Omega_1$ . Second, the ability to exploit this information in a trading strategy and finally that the benchmark for testing if the efficient market hypothesis holds is measured in economic profits. Jegadeesh and Titman (1993) found that trading strategies that bought past winners and sold past losers realized significant abnormal returns. Richardson (1993) showed that the

patterns in serial-correlation estimates and their magnitude observed in previous studies should be expected under the null hypothesis of serial independence.

Fama (1998) surveys the considerable body of empirical work on “event studies” that seeks to determine if stock prices respond efficiently to information. The “events” include such announcements as earnings surprises, stock splits, dividend actions, mergers, new exchange listings, and initial public offerings. Fama found that deceptive under reaction to information is as common as overreaction, and post-event continuation of abnormal returns is as frequent as post-event reversals. He also shows that many of the return “anomalies” arise only in the context of particular model, and that the results tend to disappear when exposed to different models for expected “normal” returns, different methods to adjust for risk, and when different statistical approaches are used to measure them. Malkiel (2003) notes that the definition of market efficiency is associated with the view that stock prices would move unpredictably. Since true news is by definition unpredictable, the resulting price changes must be unpredictable and random. On the other hand, if the full impact of an important news announcement is only grasped over a period of time, successive price changes will tend to be positively auto correlated. He examined the attacks on the EHM and concludes that stock markets are far more efficient and far less predictable. Lee et al. (2010) investigated the stationarity of real stock prices for 32 developed and 26 developing countries for the period January 1999 to May 2007 and concluded that stock markets are inefficient.

Though it is generally believed that Asian markets are less efficient, the empirical evidence does not always support this thought. The results for Asian Markets in literature are found to be mixed and subjective too. Emerging stock markets have recently attracted increasing attention from both researchers and investors especially post financial crisis. Besides its remarkable growth, emerging markets are attractive because of their low correlation with major developed stock markets, and also stock returns in many emerging markets are obvious and more predictable than developed stock markets because of unveiling systematic patterns. In many developing countries, studies have focused on predictability of return behaviour and major of the studies stresses on examining the validity of random walk hypothesis in the emerging stock markets. Most of the emerging stock markets studies have been demonstrated them to be inefficient even in the weak sense, while others found it to be efficient.

Chan. et al. (1992), found weak form efficient Asian Markets and U.S. markets, individually and jointly in the long run. Dickinson and Muragu (1994) provide substantiation of weak form of market efficiency in Nairobi Stock Exchange. Huang (1995) examine efficiency of nine Asian stock markets: Hong Kong, Indonesia, Japan, Korea, Malaysia, Philippines, Singapore, Thailand and Taiwan by using the variance ratio statistic with both assumptions homoscedastic and heteroscedastic. The hypothesis for markets of Korea and Malaysia is rejected for all holding periods, whereas the hypothesis for the Hong Kong, Singapore, and

Thailand markets is also rejected but in using the heteroscedasticity-consistent variance ratio estimator. Sharma and Kennedy (1977) compared the behaviour of stock indices of the Bombay, London and New York stock exchanges during 1963-73 using run test and spectral analysis. Both runs test and spectral analysis confirmed the random movement of stock indices for all the three stock exchanges and concluded that stocks on the BSE (Bombay Stock Exchange) follow random walk and are weak- form efficient. Ramachandran (1986) tested for the weak form efficiency of Indian Stock Markets using weekend prices of 60 scrips over the period 1976-81. Yalawar (1988) studied the month end closing prices of 122 stocks listed on the BSE during the period 1963-82 using the non-parametric tests i.e. Spearman's rank correlation test and found the behaviour of stock prices to be random i.e. weak form efficient. Khan et al. (2011) investigated the weak form market efficiency of Indian Capital Market based on the indices of two major stock exchanges of India viz. Nifty and Sensex whereas Sharma et al. (2009) examined the weak-form efficiency of eleven (11) securities listed on the BSE using weekly data from July 2007 to October 2007 by employing runs test and auto-correlation tests. The study concluded that the BSE is weak-form efficient and the stock prices are having very crimping effect on future prices which implies that an investor cannot reap out abnormal profits as the current share prices already reflect the effect of past share prices.

Srinivasan (2010) examined the random walk hypothesis to determine the validity of weak-form efficiency for two major stock markets in India by using daily observations over the span from 1st July 1997 to 31st August 2010, comprising a total of 3244 observations. The random walk hypothesis is examined using unit root tests namely, Augmented Dickey-Fuller (1979) test and the Phillips-Perron (1988) test and concluded that the null hypothesis of unit root is convincingly rejected in the case of stock market returns of two major indices, viz. Nifty and Sensex. The empirical results do not support the validity of weak-form efficiency for stock market returns of Indian stock exchanges. Venkatesan (2010) investigated the random walk behaviour of stock market returns in India and suggested that the Indian stock market is found to be efficient and supports the random walk behaviour. Sapate and Ansari (2011) concluded that the stock market returns follow random walk and they support the weak form of market efficiency. Laurence (1986) tested the weak-form efficiency in the Kuala Lumpur Stock Exchange (KLSE) and the Stock Exchange of Singapore (SES), using both auto-correlation and run tests and concluded that both markets are not efficient in the weak-form. On the contrary, a study conducted by Barnes (1986) on the Kuala Lumpur Stock Exchange (KLSE) concluded that the market is efficient in the weak-form. Urrutia (1995) employs both variance ratio of Lo and MacKinlay (1988) and runs tests to investigate random walk for the four Latin American emerging markets and concluded that the four Latin American emerging stock markets are weak- form efficient. Furthermore, Lian and Leng

(1994) examined the weak form efficiency of the KLSE employing run tests, serial correlation test, modified Box- Pierce, Q test and Van Neumann's ratio test to analyse composite index, sectoral indices and newly constructed man board all-share Emas index. The results of different statistical tests suggested that KLSE is weak form efficient with respect to monthly data. Chang et al. (1996) tested the weak form of the EMH using monthly data on the Taiwan stock exchange from 1967 to 1993 and observed that the Taiwan stock market is weak-form efficient by employing the Ljung-Box Q, the runs and the unit root tests. Using the variance ratio test, Chang and Ting (2000) examined the validity of weak form efficiency of the Taiwan stock market for the period 1971-1996 and conform to the findings of Chang et al. (1996).

Chakraborty (2006) investigated the weak-form efficiency of the Pakistan stock market using daily closing prices from January 1st 1996 to 31st December 2000. Employing variance ratio and serial correlation tests, random walk hypothesis is rejected. A study by Khawja and Mian (2005) found out that in Pakistan Stock Market, brokers can scheme to artificially raise prices and attract positive-feedback trades. Manipulation is another big issue in Pakistan Stock Market whereas the work done by Haque et al. (2011) supported that the Karachi Stock Exchange (KSE) of Pakistan is not weak form efficient because no evidence is found through the results wherein by taking higher risk the investors are rewarded. While in the study of Rashid and Husain (2009) it is indicated that the KSE of Pakistan is weak form efficient. Rabbani, Kamal and Salim (2013) tested the weak-form efficiency for KSE-100 from 1999 to 2012 by employing four tests (Augmented Dickey-fuller test, Auto-correlation function test, Phillip-perron test and Runs test) to analyze the data. All these tests except the Runs test rejected the EMH. However, they suggested market efficiency for only these two sub periods 1999-2001 and 2005-2007.

Hussain et al. (2011) studied the day of the week effect as a stock market anomaly on the equity market practices in Pakistan and concluded that Tuesday returns are quite significant and positive which implies that there exists day effect in Pakistan stock market. On one hand, Liu et al. (1997), Laurence et al. (1997), Long et al. (1999) and Lima and Tabak (2004) concluded that the Chinese markets are efficient as their respective stock returns do not exhibit predictable patterns. Wu (1996) used the serial correlation test on 8 and 12 individual shares for the period from June 1992 through December 1993 for the Shanghai and Shenzhen markets respectively and argued that CSMs seems weak. Xinpeng et al (2010) have investigated market efficiency for Shanghai and Shenzhen with unit root test. The result of the study show that data series are non-stationary, hence Chinese stock market is weak form efficient market. Liu et al. (1997) implemented unit root test using daily data from 1992 until 1995. On the basis of the results, it was asserted that both the Shanghai and Shenzhen markets are efficient.

Extant Literature supports both the view that is markets are efficient weak form and markets are not efficient in weak form. However majority of literature is of the view that markets are efficient in weak form. Thus these obvious and vacuous viewpoints in respect of weak form efficiency of market raise our interest to explore this phenomenon more deeply by applying general regression techniques thoroughly over Asian stock markets.

### **Objectives**

The primary objective of our paper is to check the efficiency of select Asian Markets in its weak form

- To verify weak form of market efficiency of five Asian Markets under study.
- To trace whether markets are efficient in weak form during different time lags in different Asian countries.
- To test whether market are behaving similarly in different time lags in different Asian countries in terms of weak form of market efficiency.

### **Data and Methodology**

As we are interested in examining weak form of efficiency in the market over different time lags so first issue is about identifying different time lags. These lags can be suitably identified with the help of graphs. Hence we shall use graphs to identify different time lags for a country. After identifying lag, we shall proceed to determine that whether a country's stock market is weak form efficient in these different time lags. For examining weak form of efficiency, it is necessary condition that past returns are not related to future returns. In other words there should exist no autocorrelation. If autocorrelation exist then it can be suitably concluded that market are not efficient in weak form. Hence our main emphasis is to find out autocorrelation in data under study.

For finding out autocorrelation in data under study we shall use general linear regression model. The main intent behind using general linear regression model is that it is most strong general test with few limitations. Hence in our general linear regression model we shall relate residuals of past returns with future return where residuals of past returns will be independent variables and residual of future return will be dependent variable. We are interested in comparison of residuals, not returns, since by taking residual we can eliminate any growth rate which is prevalent during that period. Secondly to further enhance our study we shall take residuals of past returns with lag four or five. In this way our model will present more accurate results. If residuals of past returns turn out to be significant for explaining residual of future return then this would be a proof of existence of autocorrelation and then we can suitably claim that stock markets are not efficient in weak form. As our study is basically a time series analysis and data is monthly adjusted closing indices of the representative stock market of concerned country. We have collected data for 14 years that is from 2000 to 2014 since scale of globalisation has increased during this period. Firstly we have determined the

representative stock market of each country and then proceeded to collect the data from representative stock marketor from Yahoo finance.

### ***How Weak Form of Stock Market Efficiency be recognized***

Stock markets can be termed efficient when all the necessary information affecting stock prices have already been incorporated in determining of stock prices. This implies that as all the factors have already been reflected and incorporated into current stock prices, no other factorcan influence the price of securities except new information. Movement in stock prices should be random as new information enters the market randomly. Thus a stock market is termed as efficient only if there is randomness in the movement of stock prices. If stock price movements are not random then there lies problem with efficient market hypothesis.

### ***Use of Regression method to examine efficiency***

In the present study we are only interested in analysing that whether past return affects future returns. For this purpose we have related residuals of past returns with future returns. Therefore, we have adjusted for two factors namely:

1. Any growth rate occurring during the period.
2. By comparing residuals we have also eliminated the average growth rate for the period under study.

To further enhance our method we have considered residuals of past returns for at least four or five lags. So in this way we are not interested in heteroscedastic assumption of regression model. Even if this assumption gets violated it will not affect the biasedness of estimated value of the parameter associated with independent variable. By applying this model we can know that whether the independent variables are significant or not, even in presence of heteroscedasticity. The presence of multicollinearity can be detected if model has higher R value but no significant independent variables. This situation would be dealt when we get higher value of R but no significant independent variable. However in our study if multicollinearity exists then it will further strengthen our study hence we should not be concerned with multicollinearity. In our study we are basically concerned with testing autocollinearity and we have tried to eliminate the autocollinearity from our model as discussed above. Even after this if there is presence of autocollinearity then it would further strengthen our case.

### ***Formulation of Hypothesis***

For the purpose of achieving above objectives, the following hypotheses and sub-Hypotheses are tested:

#### ***Primary Hypothesis***

1. Null Hypothesis:  $H_{0p1}$ : Asian stock markets are efficient in weak form.  
Alternate Hypothesis:  $H_{1p1}$ : Asian stock markets are not efficient in weak form.

#### ***Secondary Hypotheses***



- (i) Null Hypothesis:  $H_{0s11}$ : Indian stock market is efficient in weak form.  
Alternate Hypothesis:  $H_{1s11}$ : Indian stock market is not efficient in weak form.
- (ii) Null Hypothesis:  $H_{0s12}$ : Pakistan stock market is efficient in weak form.  
Alternate Hypothesis:  $H_{1s12}$ : Pakistan stock market is not efficient in weak form.
- (iii) Null Hypothesis:  $H_{0s13}$ : Japanese stock market is efficient in weak form.  
Alternate Hypothesis:  $H_{1s13}$ : Japanese stock market is not efficient in weak form.
- (iv) Null Hypothesis:  $H_{0s14}$ : Chinese stock markets are efficient in weak form.  
Alternate Hypothesis:  $H_{1s14}$ : Chinese stock markets are not efficient in weak form.
- (v) Null Hypothesis:  $H_{0s15}$ : Malaysian stock market is efficient in weak form.  
Alternate Hypothesis:  $H_{1s15}$ : Malaysian stock market is not efficient in weak form.

**Primary Hypothesis**

- 2. Null Hypothesis:  $H_{0p2}$ : Asian stock markets are efficient in weak form in different time lags.  
Alternate Hypothesis:  $H_{1p2}$ : Asian stock markets are not efficient in weak form in different time lags.

**Secondary Hypotheses**

- (i) Null Hypothesis:  $H_{0s21}$ : Indian stock market is efficient in weak form in different time lags.  
Alternate Hypothesis:  $H_{1s21}$ : Indian stock market is not efficient in weak form in different time lags.
- (ii) Null Hypothesis:  $H_{0s22}$ : Pakistan stock market is efficient in weak form in different time lags.  
Alternate Hypothesis:  $H_{1s22}$ : Pakistan stock market is not efficient in weak form in different time lags.
- (iii) Null Hypothesis:  $H_{0s23}$ : Japanese stock market is efficient in weak form in different time lags.  
Alternate Hypothesis:  $H_{1s23}$ : Japanese stock market is not efficient in weak form in different time lags.
- (iv) Null Hypothesis:  $H_{0s24}$ : Chinese stock markets are efficient in weak form in different time lags.  
Alternate Hypothesis:  $H_{1s24}$ : Chinese stock markets are not efficient in weak form in different time lags.
- (v) Null Hypothesis:  $H_{0s25}$ : Malaysian stock market is efficient in weak form in different time lags.  
Alternate Hypothesis:  $H_{1s25}$ : Malaysian stock market is not efficient in weak form in different time lags.

### **Primary Hypothesis**

3. Null Hypothesis:  $H_{0p3}$ : Asian stock markets behave similar in different time lags in terms of weak form of market efficiency.

Alternate Hypothesis:  $H_{1p3}$ : Asian stock markets do not behave similar in different time lags in terms of weak form of market efficiency.

### **Secondary Hypotheses**

- (i) Null Hypothesis:  $H_{0s31}$ : Indian stock market behaves similar in different time lags in terms of weak form of market efficiency.

Alternate Hypothesis:  $H_{1s31}$ : Indian stock market does not behave similar in different time lags in terms of weak form of market efficiency.

- (ii) Null Hypothesis:  $H_{0s32}$ : Pakistan stock market behaves similar in different time lags in terms of weak form of market efficiency.

Alternate Hypothesis:  $H_{1s32}$ : Pakistan stock market does not behave similar in different time lags in terms of weak form of market efficiency.

- (iii) Null Hypothesis:  $H_{0s33}$ : Japanese stock market behaves similar in different time lags in terms of weak form of market efficiency.

Alternate Hypothesis:  $H_{1s33}$ : Japanese stock market does not behave similar in different time lags in terms of weak form of market efficiency.

- (iv) Null Hypothesis:  $H_{0s34}$ : Chinese stock market behaves similar in different time lags in terms of weak form of market efficiency.

Alternate Hypothesis:  $H_{1s34}$ : Chinese stock market does not behave similar in different time lags in terms of weak form of market efficiency.

- (v) Null Hypothesis:  $H_{0s35}$ : Malaysian stock market behaves similar in different time lags in terms of weak form of market efficiency.

Alternate Hypothesis:  $H_{1s35}$ : Malaysian stock market does not behave similar in different time lags in terms of weak form of market efficiency.

### **Model in General Form and in the Form of Estimated Equation**

In weak form of efficiency future prices cannot be predicted from past prices. Hence in our analysis we will estimate a function that can be used to predict future prices from past prices. Our function is as follows:

$$\text{Residual}(t) = F [\text{Residual}(t-1), \text{Residual}(t-2), \text{Residual}(t-3), \text{Residual}(t-4), \text{Residual}(t-5)]$$

*Residual (t) is calculated as Return at t period – Average Return during the period under study*

*Return at t period is calculated as natural log (price at t/price at t-1)*

Average return is estimated by determining instantaneous growth rate of period under study whereas instantaneous growth rate is slope coefficient of natural logs of prices under study.

### ***Model in Form of Estimated Equation***

To use the above general model of regression analysis and empirical testing we have to estimate this model in the form of an equation.

$$RSD_t = b_0 + b_1 RSD_{t-1} + b_2 RSD_{t-2} + b_3 RSD_{t-3} + b_4 RSD_{t-4} + b_5 RSD_{t-5}$$

**Table: 1 Legend for Estimated Equation**

| Identifiers | Explanation                                                                                                                                        |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| $RSD_t$     | Residual for the time period t                                                                                                                     |
| $RSD_{t-1}$ | Residual for the time period with lag 1                                                                                                            |
| $RSD_{t-2}$ | Residual for the time period with lag 2                                                                                                            |
| $RSD_{t-3}$ | Residual for the time period with lag 3                                                                                                            |
| $RSD_{t-4}$ | Residual for the time period with lag 4                                                                                                            |
| $RSD_{t-5}$ | Residual for the time period with lag 5                                                                                                            |
| $b_0$       | Constant term                                                                                                                                      |
| $b_1$       | Effect of per unit change in residual of time lag 1 on residual of current time period when other residual variables of various lags are constant. |
| $b_2$       | Effect of per unit change in residual of time lag 2 on residual of current time period when other residual variables of various lags are constant. |
| $b_3$       | Effect of per unit change in residual of time lag 3 on residual of current time period when other residual variables of various lags are constant. |
| $b_4$       | Effect of per unit change in residual of time lag 4 on residual of current time period when other residual variables of various lags are constant. |
| $b_5$       | Effect of per unit change in residual of time lag 5 on residual of current time period when other residual variables of various lags are constant. |

### ***Time Series Analysis***

For the purpose of study we have selected following major stock market indices of five Asian countries:

- (i) BSE 30 as representative of Indian Stock Market.
- (ii) KSE100 as representative of Pakistani Stock Market.
- (iii) SZSE Composite as representative of Chinese Stock Market.
- (iv) NIKKEI 225 as representative of Japanese Stock Market.
- (v) KLSE Composite as representative of Malaysian Stock Market.

The five Asian countries were selected for analysis on the basis of GDP at purchasing power parity 2010. China and Japan are top two countries in North East Asia whereas India and Pakistan are top two countries in South Asia. Also the selected countries are amongst top 12 in Asia in terms of GDP at purchasing power parity. China ranked first, India second and Japan third in terms of GDP at purchasing power parity in 2010 amongst Asian countries.<sup>3</sup>

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<sup>3</sup> [http://en.wikipedia.org/wiki/List\\_of\\_Asian\\_and\\_Pacific\\_countries\\_by\\_GDP\\_%28PPP%29](http://en.wikipedia.org/wiki/List_of_Asian_and_Pacific_countries_by_GDP_%28PPP%29)

### ***Sample Size and Data Collection***

The sample includes monthly closing price of individual indices for 14 years i.e. from 2000-2014 of five Asian countries: India (Bombay Stock Exchange)<sup>4</sup>, Japan (Tokyo Stock Exchange)<sup>5</sup>, China (Shenzhen<sup>6</sup> Stock Exchange), Pakistan (Karachi Stock Exchange)<sup>7</sup> and Malaysia (Kualalumpur Stock Exchange)<sup>8</sup>, which has been collected from yahoo finance or their Stock Exchange websites. The present study is conducted using monthly closing prices of selected stock markets for a period of 14 years i.e. from 1st April 2000 to 30th September 2014. The period beginning 2000 is considered in the study as although the WTO regime was initiated in 1995 its gradual impact was witnessed only after year 2000. During this phase, the financial markets started to integrate at global level. Many new financial instruments such as ADRs, GDRs, Euro bonds etc. were introduced during this period. Firms besides diversifying its operations within domestic borders also started to diversify its ownership across borders. Thus the firms also started to get their securities listed globally. As securities were now listed at global level more efficiency was automatically expected. Also the corporate governance framework strengthened during this period since securities were listed at global level and as a consequence investors were demanding more concrete framework of corporate governance.

### ***Identification of Breaks***

After obtaining the data we plot the prices of indices with respect to time so as to identify different breaks of each country under study. Breaks have been identified with the help of abnormal movement in the stock prices whether upward or downward. If upward growth stop suddenly and down movement starts then period of continuous upward growth would be a break which would finish at the period where downward movement starts and vice versa. After identifying the breaks we studied the weak form of market efficiency within each phase of stock market.

### ***Estimation***

The data collected together with identified breaks is used to estimate the model and determine that whether the markets under the study are weak form efficient or not during different breaks.

### ***Empirical Analysis and Results***

The data collected empirically has been analysed using model discussed in the methodology. The model is then run on SPSS for linear regression, to determine the value of parameters of associated lagged time variables.

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<sup>4</sup><http://www.bseindia.com/indices/indexarchivedata.aspx>

<sup>5</sup><http://indexes.nikkei.co.jp/en/nkave/archives/data>

<sup>6</sup><http://finance.yahoo.com/q?s=399106.SZ>

<sup>7</sup><http://in.finance.yahoo.com/q?s=%5EKSE>

<sup>8</sup><http://finance.yahoo.com/q?s=%5EKLSE>

### ***Criteria for Analysis of Parameters***

To analyse whether the impact of explanatory variables is significant or not, a criteria is chosen. Any independent variable is significant or not for explaining the dependent variable depends upon the value of parameter estimated for that independent variable. To analyse the significance of value of parameter in our study we have adopted the following criteria

- i) P – Value associated with parameter
- ii) Magnitude of Parameter

#### ***P – Value Associated with Parameter***

P – Value tells us about the significance level of the parameter. However they are inversely related that is if the p – value of parameter is low then significance level is high and vice versa. In our study we have determined the cut – off for p –value to be 0.05. If p – value is less than or equal to 0.05 then we have considered it. However if p – value is more than 0.05 then we have ignored it.

#### ***Magnitude of Parameter***

Magnitude of a parameter tells about the impact of independent variable on dependent variable. If the value of parameter is high then the impact of independent variable on dependent variable is high. On the other hand if the value of parameter is low then the impact of independent variable on the dependent variable is low. In our study we have determined the cut – off value of the magnitude of the parameter. If the value of parameter is more than 0.5 then we have considered it. On the other hand if value of parameter is less than 0.5 then we have ignored it. Any independent variable is relevant or not depends upon whether P-value associated with parameter and magnitude of parameter. The relevancy of value of parameter is dependent upon the fulfilment of both of the conditions specified above. In the present study P- value considered is 0.05 which is very less in social science and also the magnitude considered is greater than 0.5. This increased the strength of our test and reduces the occurrence of type-I Error. The following 2x2 matrix table shall explain the basis for classifying parameter into relevant or irrelevant parameter.

**Table:2 Matrix Table**

| Magnitude<br>of<br>Parameter<br><br>P – value<br>of Parameter | Less than 0.5        | Equal to or more than 0.5 |
|---------------------------------------------------------------|----------------------|---------------------------|
|                                                               | Irrelevant Parameter | Relevant Parameter        |
| Equal to or less than 0.05                                    | Irrelevant Parameter | Relevant Parameter        |
| More than 0.05                                                | Irrelevant Parameter | Irrelevant                |

### ***Regression Analysis of Asian Stock Exchanges***

For determining the efficiencies in stock markets the total time period is divided into different time periods so as to achieve better results. Period include many ups and downs such as period of boom, crisis, recovery, etc. It is not possible to get proper results if time period of study is not divided into suitable breaks since markets react differently in different time frame.

#### **(i) INDIA**

We have identified the following three phases in India on the basis of stock prices movements:

- Pre-crisis period (April 2000 – December 2007)
- During crisis period (January 2008 – February 2009)
- Post crisis period (March 2009 –September 2014)

Separate regression analysis was conducted for these different time periods and the results are as follows:

**Table: 3 Summary of Indian stock markets during different periods**

| Period        | R    | R Square | Adjusted R Square | Std. Error of the Estimate |
|---------------|------|----------|-------------------|----------------------------|
| Pre-Crisis    | .598 | .358     | .318              | .00493068                  |
| During Crisis | .823 | .677     | .516              | .00564593                  |
| Post Crisis   | .555 | .308     | .246              | .00272225                  |

Source: Statistical testing

**Table: 4Value of parameters, t-statistics and P value of India during different periods**

| Period        |                           | Value of parameter | Std. Error of the value of parameter | t-statistics | P value |
|---------------|---------------------------|--------------------|--------------------------------------|--------------|---------|
| Pre-Crisis    | b <sub>0</sub> (Constant) | -.008              | .003                                 | -3.203       | .002    |
|               | b <sub>1</sub> (lag 1)    | .715               | .109                                 | 6.551        | .000    |
|               | b <sub>2</sub> (lag 2)    | -.413              | .130                                 | -3.169       | .002    |
|               | b <sub>3</sub> (lag 3)    | .326               | .135                                 | 2.422        | .018    |
|               | b <sub>4</sub> (lag 4)    | -.306              | .132                                 | -2.317       | .023    |
|               | b <sub>5</sub> (lag 5)    | .188               | .109                                 | 1.719        | .089    |
| During Crisis | b <sub>0</sub> (Constant) | .049               | .032                                 | 1.506        | .183    |
|               | b <sub>1</sub>            | .640               | .342                                 | 1.869        | .111    |
|               | b <sub>2</sub>            | -.902              | .267                                 | -3.375       | .015    |
|               | b <sub>3</sub>            | .376               | .363                                 | 1.035        | .341    |
| Post Crisis   | b <sub>0</sub> (Constant) | -.003              |                                      | -3.532       | .001    |
|               | b <sub>1</sub>            | .551               | .555                                 | 4.150        | .000    |
|               | b <sub>2</sub>            | -.541              | -.547                                | -3.725       | .000    |
|               | b <sub>3</sub>            | .524               | .572                                 | 3.668        | .001    |

|  |                |       |       |        |      |
|--|----------------|-------|-------|--------|------|
|  | b <sub>4</sub> | -.294 | -.412 | -2.239 | .029 |
|  | b <sub>5</sub> | .130  | .195  | 1.269  | .210 |

Source: Statistical testing

### ***Pre- Crisis Period***

It can be seen from table that stock market return in any month is dependent upon previous month with magnitude of 0.715. If the return in the stock market at any time is highly dependent upon any of the previous time periods then such markets are not efficient even in weak form. Another inefficiency which is noted here is about the order of occurrence of sign of parameters in an exactly alternative way (i.e. first positive than negative). Hence it can be deduced that India markets were not efficient during pre-crisis period.

### ***Crisis period***

From the above table it can be found that stock market return in any month in the stock market is highly dependent upon return with time period of lag 2 but in negative direction. The magnitude is also very high (that is 0.902). Hence, assumption of weak form of efficiency is violated during crisis period also.

### ***Post Crisis Period***

Stock market return in any time period depends upon return of previous time periods with lag 1, lag 2 and lag 3 respectively. Hence in post – crisis period also market are not efficient even in weak form.

### ***Overall Analysis of India Market***

The empirical analysis of Indian stock market shows that it is not efficient during all time periods. This clearly tells us about the inefficiency of Indian stock markets. It becomes evident that in Indian stock markets arbitrage process is not working and as a consequence anybody having large fund in their possession can dominate Indian stock markets. For the very same reasoning Foreign Institutional Investors (FII) are able to dominate Indian market. They can take stock market up to very high level as seen during pre- crisis period and they can also take it down to very low level as observed during crisis. If markets are not efficient even in weak form then it is highly unreliable and risky markets.

### **(ii) Pakistan**

We have identified following three time periods in Pakistan:

- Pre- crisis period (April 2000 – March 2008)
- During crisis period (April 2008 –February 2009)
- Post crisis period (March 2009 –September 2014)

Separate regression analysis for different time periods has been conducted and the results are as follows:

**Table:5 Summary of Pakistan Stock market**

| Period        | R    | R Square | Adjusted R Square | Std. Error of the Estimate |
|---------------|------|----------|-------------------|----------------------------|
| Pre- Crisis   | .134 | .018     | -.041             | .01018915                  |
| During Crisis | .746 | .556     | .112              | .01826939                  |
| Post Crisis   | .270 | .073     | -.011             | .00502973                  |

Source: Statistical testing

**Table: 6Value of parameters, t-statistics and P value of Pakistan**

| Period        | Model                     | Value of parameter | Std. Error of the value of parameter | t statistics | P value |
|---------------|---------------------------|--------------------|--------------------------------------|--------------|---------|
| Pre- Crisis   | b <sub>0</sub> (Constant) | -.032              | .007                                 | -4.259       | .000    |
|               | b <sub>1</sub> (lag 1)    | -.082              | .109                                 | -.755        | .452    |
|               | b <sub>2</sub> (lag 2)    | -.087              | .109                                 | -.798        | .427    |
|               | b <sub>3</sub> (lag 3)    | .000               | .109                                 | -.003        | .997    |
|               | b <sub>4</sub> (lag 4)    | -.043              | .109                                 | -.398        | .692    |
|               | b <sub>5</sub> (lag 5)    | .051               | .103                                 | .498         | .620    |
| During Crisis | (Constant)                | .282               | .119                                 | 2.381        | .098    |
|               | lag1                      | -.450              | .463                                 | -.973        | .402    |
|               | lag2                      | -.841              | .451                                 | -1.865       | .159    |
|               | lag3                      | -1.071             | .865                                 | -1.238       | .304    |
| Post Crisis   | (Constant)                | -.027              | .007                                 | -4.051       | .000    |
|               | lag1                      | -.092              | .130                                 | -.710        | .481    |
|               | lag2                      | -.239              | .131                                 | -1.825       | .073    |
|               | lag3                      | .000               | .135                                 | .001         | .999    |
|               | lag4                      | -.138              | .130                                 | -1.060       | .294    |
|               | lag5                      | .033               | .131                                 | .251         | .803    |

Source: Statistical testing

### ***Pre-Crisis period***

It can be seen from above table that stock market return in any month is independent of previous month with magnitude of less than 0.5 and p value less than 0.05. If the return in the stock market in any time is independent of the previous time periods then such markets are efficient in weak form. Another efficiency that is noted here is that there is no order in the occurrence of sign of parameters. Hence it can be deduced that Pakistan markets were efficient during pre-crisis period.

### ***Crisis period***

In this case R value is on higher side but there are no significant independent variables. This is case of multi collinearity. In this case any of the independent variable may be significant but we cannot detect that due to presence of multi collinearity. To detect the significant



variable we dropped out residual of lag1 from our model and there after we obtained the following results.

**Table: 7 Summary of Pakistan Stock market during Crisis Period**

| R    | R Square | Adjusted R Square | Std. Error of the Estimate |
|------|----------|-------------------|----------------------------|
| .645 | .416     | .124              | .01815                     |

Source: Statistical testing

**Table 8: Value of parameters, t-statistics and P value of Pakistan during Crisis period**

| Model      | Unstandardized Coefficients |            | Standardized Coefficients | t value | Sig. |
|------------|-----------------------------|------------|---------------------------|---------|------|
|            | B                           | Std. Error | Beta                      |         |      |
| (Constant) | .205                        | .087       |                           | 2.349   | .079 |
| lag2       | -.703                       | .425       | -.679                     | -1.653  | .174 |
| lag3       | -.726                       | .784       | -.381                     | -.927   | .407 |

Source: Statistical testing

Now it can be seen that although R square value has dropped but P value of lag 2 has become relatively significant (although not very high) hence it can be deduced that Pakistan markets were not highly efficient during crisis period.

#### ***Post Crisis period***

Return in any time period does not depend upon return of previous time periods. Hence in post – crisis period also market are efficient even in weak form.

#### ***Overall Analysis of Pakistan Market***

Pakistan stock market is efficient in mostly all time periods. This clearly tells us that Pakistan stock market is efficient in weak form.

#### **(iii) CHINA**

We have identified following four time periods in China:

- April 2000 – June 2005
- Pre -crisis period (July 2005 – August 2007)
- During crisis period (September 2007 – September 2008)
- Post crisis period (October 2008 – September 2014)

We have conducted separate regression analysis for these different time periods and the results are as follows:

**Table: 9Summary of Shenzhen Stock Market**

| Period | R | R Square | Adjusted R Square | Std. Error of the Estimate |
|--------|---|----------|-------------------|----------------------------|
|        |   |          |                   |                            |

|                        |       |       |       |           |
|------------------------|-------|-------|-------|-----------|
| April 2000 – June 2005 | .189  | .036  | -.059 | .01060949 |
| Pre-Crisis             | .473  | .224  | -.053 | .01274155 |
| Duringcrisis           | 1.000 | 1.000 | 1.000 | 0E-8      |
| Post Crisis            | .288  | .083  | .007  | .01098936 |

Source: Statistical testing

**Table: 10**Value of parameters, t-statistics and P value of Shenzhen stock exchange

| Period                 | Model                     | Value of parameter | Std. Error of the value of parameter | t-statistics  | P value |
|------------------------|---------------------------|--------------------|--------------------------------------|---------------|---------|
| April 2000 – June 2005 | b <sub>0</sub> (Constant) | .014               | .004                                 | 3.574         | .001    |
|                        | b <sub>1</sub> (lag 1)    | -.093              | .139                                 | -.667         | .508    |
|                        | b <sub>2</sub> (lag 2)    | .022               | .140                                 | .156          | .876    |
|                        | b <sub>3</sub> (lag 3)    | -.111              | .140                                 | -.792         | .432    |
|                        | b <sub>4</sub> (lag 4)    | -.066              | .143                                 | -.465         | .644    |
|                        | b <sub>5</sub> (lag 5)    | -.114              | .146                                 | -.778         | .440    |
| Pre –Crisis            | b <sub>0</sub> (Constant) | -.055              | .028                                 | -1.940        | .073    |
|                        | b <sub>1</sub> (lag 1)    | .286               | .272                                 | 1.053         | .310    |
|                        | b <sub>2</sub> (lag 2)    | -.501              | .301                                 | -1.665        | .118    |
|                        | b <sub>3</sub> (lag 3)    | .504               | .357                                 | 1.411         | .180    |
|                        | b <sub>4</sub> (lag 4)    | -.218              | .374                                 | -.582         | .570    |
|                        | b <sub>5</sub> (lag 5)    | -.010              | .329                                 | -.029         | .977    |
| During Crisis          | b <sub>0</sub> (Constant) | -.236              | .000                                 | -80324940.931 | .000    |
|                        | b <sub>1</sub> (lag 1)    | 1.000              | .000                                 | 102510483.027 | .000    |
| Post Crisis            | b <sub>0</sub> (Constant) | -.001              | .001                                 | -.705         | .483    |
|                        | b <sub>1</sub> (lag 1)    | -.068              | .125                                 | -.542         | .590    |
|                        | b <sub>2</sub> (lag 2)    | -.086              | .119                                 | -.722         | .473    |
|                        | b <sub>3</sub> (lag 3)    | .047               | .118                                 | .400          | .690    |
|                        | b <sub>4</sub> (lag 4)    | .239               | .118                                 | 2.025         | .047    |
|                        | b <sub>5</sub> (lag 5)    | -.034              | .121                                 | -.285         | .777    |

Source: Statistical testing

***Period April 2000 – June 2005***

It can be seen from above that return in any month in the stock market is independent of previous month with magnitude of less than 0.5 and P value greater than 0.05. If the return in the stock market in any time is highly independent of the previous time periods then such markets are efficient in weak form. The order of occurrence of sign of parameters is not

following any pattern. Hence it can be deduced that Shenzhen markets were efficient during this period.

#### ***Pre- crisis period***

It can be seen from above table that return in any month in the stock market is independent of previous month because both the conditions are not satisfied. If the return in the stock market in any time is highly independent of the previous time periods then such markets are efficient in weak form. Hence it can be deduced that Shenzhen markets were efficient during pre-crisis period.

#### ***Crisis period***

It is the perfect case of inefficiency since all the conditions are perfectly met. The conditions are:

- i) R square is equal to one
- ii) Value of the parameter associated with residual of lag1 is equal to one
- iii) P value is equal to zero

Hence Shenzhen market is inefficient in weak form during-crisis period.

#### ***Post crisis period***

Shenzhen markets were efficient during post-crisis period as return in any month in the stock market is independent of previous month with magnitude of less than 0.5 and p value greater than .05. Moreover there is no order of occurrence of signs also. Hence Shenzhen markets were efficient during post-crisis period

#### ***Overall Analysis of Shenzhen Market***

Shenzhen stock market is efficient during all time periods except during crisis period and during crisis we observed that there exists perfect case of inefficiency.

#### **(iv) Malaysia**

We have identified following three time periods in Kuala Lumpur:

- *Pre - crisis period (April 2000 –September 2007)*
- *During crisis period (October 2007 – September 2008)*
- *Post crisis period (October 2008 – September 2014)*

We have conducted separate regression analysis for these different time periods and the results are as follows:

**Table:11 Summary of Malaysian Market**

| Period        | R    | R Square | Adjusted R Square | Std. Error of the Estimate |
|---------------|------|----------|-------------------|----------------------------|
| Pre-Crisis    | .241 | .058     | -.002             | .00699145                  |
| During Crisis | .352 | .124     | -.533             | .00690366                  |
| Post Crisis   | .161 | .026     | -.055             | .0044215                   |

Source: Statistical testing

**Table: 12 Value of parameters, t-statistics and P value of Malaysian Markets**

| Period        | Model                     | Value of parameter | Std. Error of the value of parameter | t-statistics | P value |
|---------------|---------------------------|--------------------|--------------------------------------|--------------|---------|
| Pre-Crisis    | b <sub>0</sub> (Constant) | -.005              | .002                                 | -3.142       | .002    |
|               | b <sub>1</sub> (lag 1)    | .071               | .107                                 | .662         | .510    |
|               | b <sub>2</sub> (lag 2)    | -.099              | .109                                 | -.911        | .365    |
|               | b <sub>3</sub> (lag 3)    | -.029              | .109                                 | -.261        | .795    |
|               | b <sub>4</sub> (lag 4)    | -.016              | .107                                 | -.154        | .878    |
|               | b <sub>5</sub> (lag 5)    | .196               | .106                                 | 1.855        | .067    |
| During Crisis | b <sub>0</sub> (Constant) | .043               | .031                                 | 1.355        | .247    |
|               | b <sub>1</sub> (lag 1)    | -.378              | .563                                 | -.672        | .539    |
|               | b <sub>2</sub> (lag 2)    | -.242              | .499                                 | -.485        | .653    |
|               | b <sub>3</sub> (lag 3)    | .004               | .470                                 | .009         | .993    |
| Post Crisis   | b <sub>0</sub> (Constant) | -.006              | .002                                 | -2.989       | .004    |
|               | b <sub>1</sub> (lag 1)    | -.015              | .128                                 | -.113        | .910    |
|               | b <sub>2</sub> (lag 2)    | .092               | .128                                 | .723         | .473    |
|               | b <sub>3</sub> (lag 3)    | .071               | .128                                 | .552         | .583    |
|               | b <sub>3</sub> (lag 4)    | -.067              | .128                                 | -.522        | .603    |
|               | b <sub>3</sub> (lag 5)    | .079               | .128                                 | .614         | .542    |

Source: Statistical testing

It is one of the cases where market is efficient during all time periods. Kuala Lumpur markets were efficient during all periods since both the conditions are not satisfied for any of the periods. Moreover there is no observed pattern for order of occurrence of signs.

#### (v) Japan

We have identified following time periods in Japan:

- April 2000 – February 2003
- Pre-crisis period (March 2003 – May 2007)
- During crisis period (June 2007 - January 2009)
- Post crisis period (February 2009 - July 2012)
- August 2012 - September 2014

We have conducted separate regression analysis for these different time periods and the results are as follows:

**Table: 13Summary of Japan**

| Period                      | R    | R Square | Adjusted R Square | Std. Error of the Estimate |
|-----------------------------|------|----------|-------------------|----------------------------|
| April 2000 to February 2003 | .289 | .083     | -.116             | .00623638                  |

|                              |      |      |       |           |
|------------------------------|------|------|-------|-----------|
| Pre-crisis period            | .307 | .094 | -.022 | .00408322 |
| During Crisis                | .545 | .297 | -.142 | .01074683 |
| Post Crisis                  | .276 | .076 | -.078 | .00661255 |
| August 2012 - September 2014 | .327 | .107 | -.212 | .00540986 |

Source: Statistical testing

**Table: 14 Value of Parameters, t–statistic and p–value of Japan Market**

| Period                       | Model                     | Value of parameter | Std. Error of the value of parameter | t-statistics | P value |
|------------------------------|---------------------------|--------------------|--------------------------------------|--------------|---------|
| April 2000 to February 2003  | b <sub>0</sub> (Constant) | .024               | .009                                 | 2.691        | .013    |
|                              | b <sub>1</sub> (lag 1)    | .049               | .205                                 | .242         | .811    |
|                              | b <sub>2</sub> (lag 2)    | -.100              | .193                                 | -.519        | .608    |
|                              | b <sub>3</sub> (lag 3)    | -.068              | .188                                 | -.361        | .721    |
|                              | b <sub>4</sub> (lag 4)    | -.233              | .190                                 | -1.228       | .232    |
|                              | b <sub>5</sub> (lag 5)    | .016               | .194                                 | .083         | .935    |
| Pre-crisis period            | b <sub>0</sub> (Constant) | -.014              | .004                                 | -4.037       | .000    |
|                              | b <sub>1</sub> (lag 1)    | -.051              | .151                                 | -.341        | .735    |
|                              | b <sub>2</sub> (lag 2)    | .140               | .149                                 | .938         | .354    |
|                              | b <sub>3</sub> (lag 3)    | .098               | .148                                 | .663         | .511    |
|                              | b <sub>4</sub> (lag 4)    | -.033              | .144                                 | -.232        | .817    |
|                              | b <sub>5</sub> (lag 5)    | -.253              | .141                                 | -1.790       | .081    |
| During Crisis                | b <sub>0</sub> (Constant) | .077               | .036                                 | 2.136        | .065    |
|                              | b <sub>1</sub> (lag 1)    | .093               | .318                                 | .294         | .776    |
|                              | b <sub>2</sub> (lag 2)    | -.323              | .318                                 | -1.016       | .339    |
|                              | b <sub>3</sub> (lag 3)    | .046               | .330                                 | .138         | .893    |
|                              | b <sub>4</sub> (lag 4)    | -.061              | .484                                 | -.127        | .902    |
|                              | b <sub>5</sub> (lag 5)    | -.870              | .540                                 | -1.611       | .146    |
| Post crisis                  | b <sub>0</sub> (Constant) | .003               | .002                                 | 2.153        | .039    |
|                              | b <sub>1</sub> (lag 1)    | -.158              | .178                                 | -.888        | .381    |
|                              | b <sub>2</sub> (lag 2)    | -.142              | .177                                 | -.803        | .428    |
|                              | b <sub>3</sub> (lag 3)    | -.044              | .182                                 | -.244        | .809    |
|                              | b <sub>4</sub> (lag 4)    | -.149              | .178                                 | -.838        | .409    |
|                              | b <sub>5</sub> (lag 5)    | -.137              | .177                                 | -.776        | .444    |
| August 2012 - September 2014 | b <sub>0</sub> (Constant) | -.017              | .009                                 | -2.003       | .065    |
|                              | b <sub>1</sub> (lag 1)    | .098               | .264                                 | .370         | .717    |
|                              | b <sub>2</sub> (lag 2)    | -.029              | .248                                 | -.116        | .909    |
|                              | b <sub>3</sub> (lag 3)    | .258               | .242                                 | 1.062        | .306    |
|                              | b <sub>4</sub> (lag 4)    | -.213              | .249                                 | -.856        | .407    |
|                              | b <sub>5</sub> (lag 5)    | .010               | .247                                 | .040         | .969    |

Source: Statistical testing

Like Kuala Lumpur Stock Market this is also one of the cases where market is efficient in all the time periods. Japan stock market is efficient during all time periods. Return in any month in the stock market is independent of previous month with magnitude of less than 0.5 and P value more than 0.05. Hence it can be deduced that Japan markets were efficient in all the periods.

### **Conclusion**

Hitherto there exists controversy regarding efficiency of market in weak form. Few researchers have proved that markets are efficient in weak form while others have proved otherwise. However, extant literature is dominated by the idea that markets are efficient in weak form. As a beginning point the study assumes that markets are efficient in weak form and then test these hypotheses by applying regression method. To test this, study relates the residuals of stock market returns of current period with residuals of return with previous periods. Only if a connection is established between current return and lag return then it can be concluded that markets are not efficient in weak form or market is not following random walk.

At last after reviewing and analysing the result of this study we arrived at following findings:

- i) Contrary to the notion that Asian Markets are efficient in weak form the findings of our study shows that Asian Markets are not efficient in weak form.
- ii) As markets are not efficient in weak form so arbitrage process is not working sufficiently.
- iii) As arbitrage process is not working sufficiently so there exists a possibility of making abnormal gains in share market.
- iv) However these abnormal gains can be achieved if one person has sufficient resources. For the similar reason FII and hedge funds are able to dominate any stock market in Asia.
- v) As markets are not efficient in weak form this implies that market would be deviating from its fundamentals. This means small investors should remain away from these markets and only invest through financial intermediaries with good credit ratings.

Our study provides empirical evidence that Asian markets are not weak form efficient. Hence we may conclude that arbitrage process is not working sufficiently and there exists a possibility of making abnormal gains in stock market by the people who have sufficient resources, for example FIIs. Since markets are not weak form efficient it implies that market would be deviating from its fundamentals. Hence small investors should remain away from these markets and are suggested to make investment through financial intermediaries with good credit rating and perhaps with strong market analysis.

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